Graph Masked Autoencoders With Relationship-Aware Learning for Hyperspectral Image Clustering

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***Abstract*—Hyperspectral image (HSI) clustering is a fundamental yet challenging task. Recently, significant advances have been achieved through methods based on graph contrastive learning and graph autoencoders. Nevertheless, these existing methods still encounter notable limitations. In particular, inappropriate graph augmentation strategies within graph contrastive learning can deteriorate the intrinsic structural and semantic information embedded in the original graph data. Meanwhile, graph autoencoders commonly suffer from representation collapse and are susceptible to noisy representations, thus restricting further improvements in clustering performance. To mitigate these limitations, this article proposes a novel jointly optimized and trainable framework named graph masked autoencoders with relationship-aware learning (GMARC) for HSI clustering, which is the first to apply masked graph autoencoders (MGAEs) to HSI clustering tasks. GMARC thoroughly exploits the spectral characteristics of HSI data and effectively models their underlying high-dimensional graph structures. In particular, within an online–target network architecture, a strategy of random masking on node features is introduced to fully capture the intrinsic structural characteristics of HSI, thereby significantly enhancing the model's generalization capability. Furthermore, we propose a relationship-aware learning module (RAL), explicitly constructing both global and local relationships to enrich data representations. This module enables the model to be actively aware of positive and negative node pairs within the graph, thereby reinforcing the feature consistency among positive pairs and reducing correlation among negative pairs. Consequently, our method achieves precise extraction of discriminative features, critical for effective clustering. Extensive experiments on three widely used datasets (Xu Zhou, Salinas, and Trento) demonstrate GMARC's superior clustering performance, achieving overall accuracies of 79.68%, 81.38%, and 90.85%, respectively, and substantially surpassing state-of-the-art approaches.**

***Index Terms*—Deep clustering, graph autoencoder, hyperspectral images (HSIs), masking, relationship-aware learning (RAL).**

Received 19 March 2026; revised 7 May 2026 and 16 June 2026; accepted 22 June 2026. Date of publication 2 July 2026; date of current version 27 July 2026. *(Corresponding authors: Xiang Yang; Dayu Hu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3709391

I. Introduction

HYPERSPECTRAL image (HSI) involves acquiring data across numerous contiguous spectral bands, wherein the spectral reflectance or radiance of each pixel encapsulates both the compositional properties and the intricate structural nuances of the object [1], [2]. Based on this abundant spectral information, HSI has been widely applied in fields such as urban planning [3], [4], [5], agricultural monitoring [6], [7], [8], environmental assessment [9], [10], [11], and crop identification [12], [13]. However, due to their high dimensionality and homogeneity, HSIs often contain significant redundant information and interference bands [14], [15], which necessitates the reliance on costly manual labeling for data processing and complicates the acquisition of high-quality pixel-level labels.

In recent years, unsupervised learning has garnered considerable attention in HSI processing [16], [17], [18]. As a result, unsupervised HSI clustering has become a key research focus within the remote sensing community. HSI clustering [19], [20], [21] leverages unlabeled data to uncover the relationships between different pixels, grouping pixels with similar semantic content into the same cluster, thereby effectively addressing the challenge of limited labeled data. Early HSI clustering approaches primarily concentrate on pixel-level spectral information [22], [23]. However, these methods typically rely on the features of individual pixels, neglecting the spatial correlations within their surrounding neighborhoods and struggling to capture global or higher order structural information of HSI. Consequently, pixel-level methods often exhibit suboptimal performance when addressing categories with complex spatial distributions. Furthermore, pixel-level clustering methods are generally hindered by high computational complexity, challenging optimization, and substantial computational overhead. To mitigate these limitations, superpixel segmentation techniques, such as simple linear iterative clustering (SLIC), have gained widespread use in HSI clustering tasks [24],

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[25]. By grouping adjacent pixels with similar properties into larger superpixel regions, these methods can more effectively capture local spatial characteristics, reduce noise interference, and enhance the representation quality of spectral information [26]. Superpixel regions demonstrate superior compactness and consistency, preserving the boundary information of target regions, which significantly improves robustness in complex scenarios [27].

To effectively aggregate long-range spatial dependencies and capture the vector-level features of superpixels, HSI clustering tasks based on superpixels have been successfully tackled using graph neural networks (GNNs). Since superpixels derived from HSI inherently possess well-defined adjacency relationships, representing them as graph nodes and connecting those with similar semantic information via edges enables the structure to exploit spectral information more effectively. In graph-structured HSI clustering, GNN-based HSI clustering methods are predominantly classified into contrastive learning-based [28], [29], [30], and autoencoder-based [31], [32] approaches. Contrastive learning-based methods typically perform data augmentation by generating abundant positive and negative sample pairs, which effectively minimizes the distances between similar samples and maximizes the separability of different samples, thereby facilitating the extraction of robust feature representations. On the contrary, the graph autoencoder-based method [33] fully exploits the topological structure of the constructed graph and aggregates the features of neighboring nodes, thereby learning deep feature representations to reconstruct the graph structure and feature information. Moreover, these approaches often integrate graph convolutional networks (GCNs) [34] and graph attention networks (GATs) [35] as encoders and decoders during training, thus capturing a more comprehensive array of data features.

Nevertheless, each of these HSI clustering approaches exhibits inherent limitations. Contrastive learning-based methods are heavily dependent on data augmentation, and devising effective augmentation strategies for graph data remains particularly challenging. These augmentation strategies may disrupt the intrinsic structure or semantic content of the original graph. Consequently, the resultant representations may not accurately capture the original characteristics of the data. For instance, randomly removing nodes can result in the loss of crucial nodes, leading to a degradation of the structural information within the graph. Similarly, random edge deletion may disrupt the internode relationships that are essential for maintaining the graph's connectivity. In addition, contrastive learning-based methods typically require the meticulous construction of negative sample pairs, and the task of selecting or generating appropriate negatives continues to pose a significant challenge, as unsuitable choices can adversely affect the representation learning process. Autoencoder-based approaches circumvent the challenges posed by contrastive learning techniques by focusing on the reconstruction of input graph data. Nevertheless, the straightforward design of graph autoencoders renders them susceptible to representation collapse during training, thereby compromising their robustness in feature extraction. Furthermore, current methodologies may inadvertently retain extraneous information that is irrelevant to clustering, consequently introducing redundant noise into the resultant feature representations.

To effectively address the aforementioned challenges, this article proposes a joint optimization training framework based on graph masked autoencoders (MAEs), termed graph masked autoencoders with relationship-aware learning (GMARC). Masked graph autoencoders (MGAEs) have demonstrated effectiveness in addressing common issues of representation collapse and redundant noise during autoencoder training. To leverage this, we innovatively integrate MGAE with relationship-aware learning (RAL) by fully exploiting the characteristics of HSI as an intrinsic driving signal for the model. This enables the collaborative optimization of masked reconstruction and relationship awareness, capturing structural information inherent in the graph at a higher level, extracting more discriminative feature representations, and significantly enhancing clustering performance. Concretely, an online–target network architecture is introduced, in which a randomized masking strategy is applied to the graph node features. Masked and unmasked graphs are separately encoded via dedicated graph encoders, yielding informative and robust latent representations. This architecture not only effectively mitigates representation collapse but also significantly enhances robustness against noise typically observed in HSI. Moreover, we impose alignment constraints on latent features corresponding to masked nodes, guiding the model to more effectively reconstruct challenging feature details. In summary, the main contributions of this article are as follows.

1) We propose a novel HSI clustering method named GMARC. This method is based on GCN architecture, introduces the masking strategy into HSI clustering for the first time, and leverages the reconstruction task to guide the model in effectively learning robust features.
2) We design a RAL module that comprehensively exploits the high-dimensional structural information in graph data. It establishes multiple relationships without altering the original graph topology, thereby enhancing the perception of connections between positive and negative sample pairs and enabling a more accurate capture of intrinsic correlations.
3) We conducted comprehensive experiments on three publicly available datasets to assess the performance of the proposed model. The results demonstrate the superiority and effectiveness of the method.

II. Related Work

A. Masked Graph Modeling

Masked language models exemplified by models like BERT [36] are extensively utilized in natural language processing. By predicting masked words, these language models gain an understanding of semantic relationships within contexts, enabling the acquisition of more profound language representations. He et al. [37] pioneered the incorporation of masking strategies into the computer vision domain through the development of the MAE. MAE randomly masks pixels of the input image, forcing the model to learn both global

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and local feature representations. The success of MAE has proven the effectiveness of large-scale masking strategies when applied to self-encoder architectures.

In recent years, research in the field of deep learning has increasingly concentrated on generative self-supervised learning. Within the domain of graph data structures, masked graph self-models (MGMs) have garnered significant attention as pivotal generative models. For instance, Tan et al. [38] introduced a tailored cross correlation decoder that models node correlations at varying granularities to reconstruct masked edge structures. GraphMAE [39] placed greater emphasis on reconstructing graph node features by employing masking strategies multiple times across different stages. This approach facilitates the learning of more meaningful node representations, thereby preventing the model from converging to trivial solutions. Hou et al. [40] incorporated regularization into the target reconstruction process to mitigate the risk of model overfitting and address susceptibility to interference from low-discriminative features. Li et al. [41] introduced path masking, which compels the model to more effectively utilize structure-dependency patterns and capture high-order proximities. In addition, it integrates a degree decoder that employs two distinct architectures during the decoding phase to restore both the edge structure and degree condition of the graph, thereby achieving a scalable model. Tian et al. [42] proposed a framework designed for heterogeneous graphs, employing multiple training strategies to capture comprehensive graph information, making it more expressive for complex graph structures. UGMAE [43] incorporates the importance of each node into the masking process, reconstructing targets based on importance ranking, and fully leveraging the semantically rich representation space for semantic learning. Luo et al. [44] applied contrastive learning within constructed multiview perspectives and combined cross information among multiple views, enabling the model to learn more discriminative graph representations.

While masking techniques remain unexplored in the field of HSI clustering, we fill this research gap by proposing a relation-aware GMAE, marking the first application of such a paradigm to HSI clustering tasks.

B. Graph-Based HSI Clustering

Graph-based HSI clustering methods demonstrate a marked advantage in capturing the intricate intrinsic structure of HSI data and have emerged as a dominant approach in this research area [45], [46], [47], [48]. Traditional clustering techniques typically depend on mathematical models and statistical methods, underpinned by explicit mathematical formulations and clearly defined optimization objectives [49]. For instance, Cai et al. [50] extended graph convolutional subspace clustering to non-Euclidean domains and employed the Frobenius norm to derive a globally optimal closed-form solution, thereby streamlining the clustering process. Similarly, Huang et al. [51] integrated graph convolution with dictionary atoms to exploit both spatial contextual and spectral neighborhood information among the atoms, effectively aggregating local neighborhood details to enhance feature representation. Furthermore, Cui et al. [20] leveraged diffusion geometry theory to characterize the inherent data structure while ensuring that the diffusion process captures both spatial and spectral characteristics, thus rendering the diffusion distance a more accurate measure of pixel similarity.

In parallel with the continuous advancements in deep learning, deep clustering approaches have garnered significant attention, leading to a surge of innovative methods [21], [52], [53]. For instance, Ding et al. [54] introduced a local-preserving low-pass graph convolution autoencoder that leverages low-pass graph convolution in conjunction with a hierarchical graph attention mechanism to derive enhanced latent representations. Similarly, Luo et al. [55] integrated an autoencoder with a Transformer architecture, employing a Siamese dual-former graph module based on superpixel features to effectively capture long-range dependencies in graph convolution features. Within the realm of graph-based deep clustering, the proven efficacy of contrastive learning has established a consensus, rendering the design of robust data augmentation strategies a fundamental objective. In particular, Guan et al. [56] constructed dual graph convolution subspaces and utilized contrastive learning to reinforce the consistency of positive samples across multiple views, thereby yielding a more discriminative affinity matrix; Qi et al. [30] exploited pretraining to comprehensively mine the features of HSI, and devised a data augmentation scheme that incorporates both sample-level alignment and cluster center-level contrast to effectively regulate interclass relationships. EMVCC [57] employs Transformer-based spectral feature extraction while formulating heterogeneous and homogeneous views as positive and negative samples, respectively, to enhance the network's ability to learn clustering-friendly representations. Moreover, to rigorously constrain interclass disparities between positive and negative samples, Yang et al. [29] adopted a multilayer perceptron for the direct generation of low-dimensional embeddings and introduced a multiboundary loss function designed to capture complementary spatial and neighborhood information. Despite notable progress achieved by the aforementioned deep clustering methods in enhancing clustering accuracy, these approaches remain insufficient in comprehensively analyzing graph structures from both global and local perspectives. Consequently, they fall short of adequately exploiting the high-order attributes and structural characteristics inherent in graph data.

III. Method

In this section, we first clarify the definition of the research problem and introduce the GMARC framework developed to address it. Thereafter, a detailed analysis of each constituent component of the framework is provided, including the graph encoder, graph decoder with remasked strategy, and RAL. Fig. 1 illustrates the architecture of the proposed GMARC.

A. Problem Definition and Notations

In this article, we introduce an end-to-end network architecture that seamlessly integrates reconstruction and RAL for clustering through the use of an MAE. Acknowledging the

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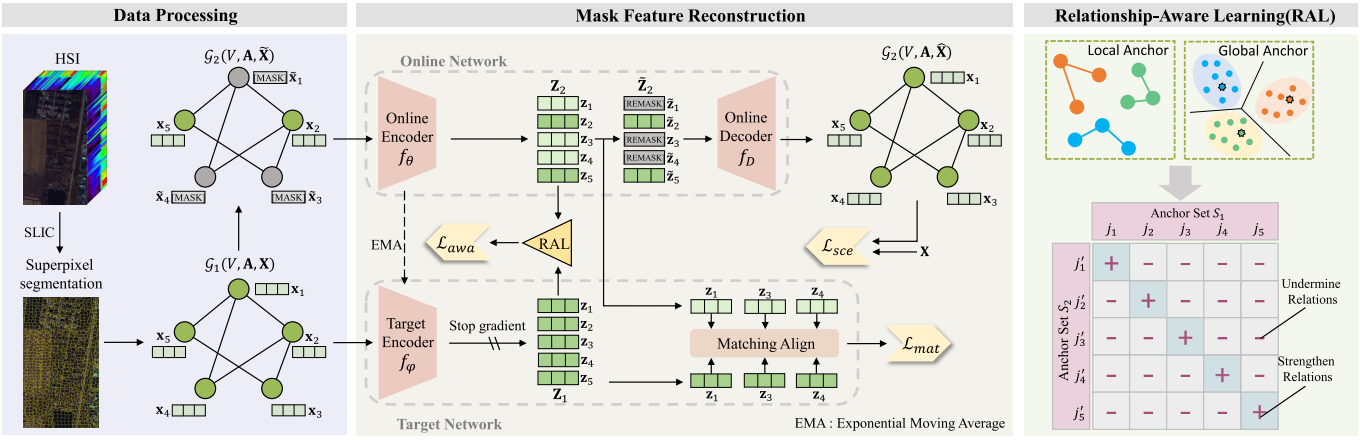


Fig. 1. Workflow of the GMARC framework. Initially, a graph constructed from superpixel segmentation results serves as the network input. The online and target networks then separately encode the masked and unmasked graphs to generate informative and robust latent representations. Subsequently, the encoded features from the masked nodes are aligned. The latent representations derived from the online network then undergo a secondary masking procedure followed by decoding to yield high-quality reconstructed features. Finally, a RAL module establishes multiple relationships to extract features with enhanced discriminability.

challenges of pixel-level optimization, we leverage superpixel segmentation techniques to significantly reduce computational complexity while ensuring the preservation of essential spectral features. In particular, for the input HSI $\mathcal{S} \in \mathbb{R}^{W\times H\times D}$, the SLIC algorithm is used to perform superpixel segmentation, resulting in the superpixel set $\mathcal{T} \in \mathbb{R}^{M}$. For each superpixel representation, we assign a label to associate it with the corresponding pixel-level representation and compute the centroid coordinates. Subsequently, based on these centroid coordinates and spectral information, for each node, we identify and establish connections with its k-nearest neighbors, followed by the application of a Gaussian kernel function for data smoothing. This process ultimately results in the construction of the feature graph $G_1 = (\mathcal{V}, \mathbf{A}, \mathbf{X})$, which serves as the primary input to our model.

1) Notation: Consider an undirected graph $G = (\mathcal{V}, \mathbf{A}, \mathbf{X})$, where $\mathcal{V}$ represents the set of nodes within the graph and M denotes the total number of nodes. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{M\times M}$ is constructed based on the similarity between nodes, such that $a_{ij} = 1$ if an edge exists between nodes v_i and v_j, and $a_{ij} = 0$ otherwise. The feature matrix $\mathbf{X} \in \mathbb{R}^{M\times d}$ contains the feature vectors for each node, with each row corresponding to a node's features, where d indicates the dimensionality of these features. The adjacency matrix is normalized as $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}$, where $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\hat{\mathbf{D}}_{ii} = \sum_{j=1}^{M} \hat{a}_{ij}$ are the degree matrix. Furthermore, T_1 denotes the number of global anchor points sampled to construct global relationships, and T_2 represents the number of highest scoring neighbor nodes selected to establish local relationships.

2) Graph Convolutional Network: GCN extends the traditional convolutional operations to data structured as a graph, aiming to efficiently aggregate information from neighboring nodes and the node itself, thereby enabling the integration and propagation of both local and global information. The core operation of a GCN is described by the subsequent equation

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{1}$$

where $\mathbf{H}^{(l)}$ represents the node feature matrix at the lth layer, $\mathbf{W}^{(l)}$ is the learnable weight matrix, and σ is the activation function applied at each layer, typically ReLU or Tanh. The resulting $\mathbf{H}^{(l+1)}$ is the updated node feature matrix after the convolution operation. By stacking multiple layers of GCN, the model is able to learn increasingly abstract semantic information, which enhances its performance on the target task.

B. Graph Encoder

In HSI clustering tasks, capturing highly discriminative and high-dimensional spectral features remains a significant challenge. The primary motivation for our graph encoder design is to overcome the limitations of traditional autoencoders, which often fall into representation collapse or learn merely superficial features. To address this and fully exploit the spatial–spectral redundancy inherent in HSI data, we propose a graph MAE that learns more expressive feature representations by reconstructing masked features. This approach not only effectively mitigates representation collapse risks but also enhances robustness against noise.

In the model training process, the graph encoder, as the primary step, aims to project the original graph into a latent space, thereby generating compact and semantically rich representations. First, we sample a subset of nodes from the node set $\mathcal{V}$ at a predetermined sampling rate ξ to form the masked node set $\tilde{\mathcal{V}}$, and assign the [MASK] token $\mathbf{x}_{[MASK]}$ to each node within this set. In practice, a learnable vector is trained to replace the features of these masked nodes. Given that GCN leverages information from neighboring nodes for reconstruction, a uniform random sampling strategy without replacement is adopted to select the masked nodes. This strategy facilitates a deeper understanding of the graph structure and enables the model to effectively extract feature information. Formally, the masked feature matrix $\tilde{\mathbf{X}}$ of the graph G_2 can be defined as

$$\tilde{\mathbf{x}}_i = \begin{cases} \mathbf{x}_{[MASK]}, & \text{if} \quad v_i \in \tilde{\mathcal{V}} \\ \mathbf{x}_i, & \text{if} \quad v_i \notin \tilde{\mathcal{V}} \end{cases} \tag{2}$$

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here, we apply the proposed strategy to the original graph G_1, generating a graph $G_2 = (\mathcal{V}, \mathbf{A}, \tilde{\mathbf{X}})$ based on the masked perspective. Then, the masked features serve as input to the online encoder.

The target encoder f_ϕ accepts the original graph G_1 as its input and utilizes a two-layer GCN for feature extraction. This encoder is structurally consistent with the online encoder, but its parameters do not directly participate in gradient backpropagation; rather, they are updated via an exponential moving average (EMA) mechanism driven by the online encoder. Its specific definition is as follows:

$$\phi \leftarrow \epsilon\phi + (1-\epsilon)\,\theta \tag{3}$$

where ϵ is a smoothing factor closer to 1, retaining certain reliable steps for updating parameters.

To assess the representational capability of the online encoder regarding reconstructed features, we propose a matching loss function, which aligns the latent embeddings $\mathbf{Z}_1$ and $\mathbf{Z}_2$ produced by the online and target encoders, respectively. It is important to note that, since feature reconstruction primarily emphasizes the recovery of the more challenging masked node features, the matching loss is computed exclusively over the nodes within the masked node set $\tilde{\mathcal{V}}$. The matching loss $\mathcal{L}_{\text{mat}}$ is defined as follows:

$$\mathcal{L}_{\text{mat}} = \frac{1}{|\tilde{\mathcal{V}}|}\sum_{v_i\in\tilde{\mathcal{V}}}\left\|\mathbf{z}_1^{(i)} - \mathbf{z}_2^{(i)}\right\|^2. \tag{4}$$

The proposed matching loss function aligns the latent representations of masked nodes, thereby ensuring more consistent output features and significantly enhancing the reconstruction of complex node characteristics. Moreover, by restricting the computation of the matching loss solely to masked nodes, the model is encouraged to concentrate on recovering challenging feature details, ultimately bolstering its representational capacity and generalization performance.

C. Graph Decoder With Remasked Strategy

Upon completion of the graph encoding phase, the second crucial component of the proposed method is the graph remasked decoder, which aims to project the latent representations produced by the encoder back into the original feature space. While the first masking step applied to the input node features creates a challenging pretext task for the encoder to extract deep structural information, we introduce a second masking operation, referred to as the remasked strategy, which is applied to the latent representations before the decoding stage. The primary rationale behind this dual-masking approach is to create a more difficult and effective reconstruction task for the decoder. Accurate reconstruction from a standard latent space might lead the model to learn superficial representations. By compelling the decoder to reconstruct original features from remasked latent representations, the model effectively prevents two major issues common in graph autoencoders: representation collapse and the oversmoothing problem. This yields an updated latent representation denoted as $\tilde{\mathbf{Z}}$, comprising individual latent vectors $\tilde{\mathbf{z}}_i$, defined formally as follows:

$$\tilde{\mathbf{z}}_i = \begin{cases}\mathbf{z}_{[REMASK]}, & \text{if} \quad v_i \in \tilde{\mathcal{V}} \\ \mathbf{z}_i, & \text{if} \quad v_i \notin \tilde{\mathcal{V}}.\end{cases} \tag{5}$$

Following the proposed methodology, graphs G_1 and G_2 are initially encoded separately, yielding their corresponding latent representations $\mathbf{Z}_1$ and $\mathbf{Z}_2$. Subsequently, a secondary masking operation is applied to the latent representation $\mathbf{Z}_2$, producing the remasked feature representation $\tilde{\mathbf{Z}}_2$. Ultimately, the resulting features $\tilde{\mathbf{Z}}_2$ are fed into a structurally symmetric GCNs decoder to generate the final decoded outputs. The above procedure can be formalized as $\hat{\mathbf{X}} = f_D(\tilde{\mathbf{Z}}_2) = \text{GCN}(\tilde{\mathbf{Z}}_2)$. This architecture effectively recovers the features of masked nodes without causing the notorious over-smoothing problem. We employ the loss function $\mathcal{L}_{sce}$ to quantify the discrepancy between reconstructed features and original node features. In addition, a scaling factor $\gamma \geq 1$ is introduced to down-weight the training contribution from easy samples, thereby encouraging the model to emphasize learning from challenging samples. The specific definition is as follows:

$$\begin{aligned}\mathcal{L}_{\text{sce}} &= \frac{1}{|\tilde{\mathcal{V}}|}\sum_{v_i\in\tilde{\mathcal{V}}}\left(1 - \text{sim}\left(\mathbf{x}_i, \hat{\mathbf{x}}_i\right)\right)^\gamma \\ &= \frac{1}{|\tilde{\mathcal{V}}|}\sum_{v_i\in\tilde{\mathcal{V}}}\left(1 - \frac{\mathbf{x}_i^T\hat{\mathbf{x}}_i}{\|\mathbf{x}_i\|_2\cdot\|\hat{\mathbf{x}}_i\|_2}\right)^\gamma, \quad \gamma \geq 1\end{aligned} \tag{6}$$

where $\text{sim}(\cdot)$ denotes the cosine similarity between two vectors and γ represents a hyperparameter introduced to modulate the distribution of loss weights among individual samples.

D. Relationship-Aware Learning

While the graph encoder effectively extracts robust node-level features, it lacks explicit constraints on the complex, higher order topological dependencies among superpixels. The motivation behind our RAL module is to explicitly capture the Non-IID nature of hyperspectral graphs, proactively guiding the model to discriminate between structurally similar and dissimilar node pairs without relying on heuristic augmentations. To intuitively illustrate this mechanism, Fig. 2 visualizes the detailed process of establishing these global and local relationships.

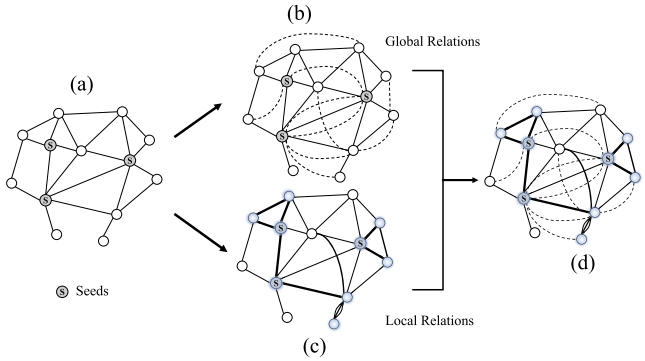


Fig. 2. Diagram of the process for establishing global and local relationships. (a) Initial graph. (b) Polynomial sampling. (c) Personalized PageRank. (d) Combined graph.

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1) Establishing Relationships: This module is designed to analyze graph structures characterized by Non-IID features. Leveraging the latent representations derived from the encoding phase, the module constructs both global and local relationships, ensuring the integrity of the graph structure is maintained. This enables a deeper exploration of the higher order attributes and structural features of the data, providing a theoretical foundation for HSI clustering tasks.

Motivated by the work of Yi et al. [58], this article proposes a polynomial sampling-based anchor node selection method to effectively extract global graph features and establish global relationships. In particular, we introduce a nonuniform sampling strategy to compute the sampling probability for each node, facilitating balanced information propagation across the graph. Formally, the weight w_i of a node v_i is defined as

$$\begin{aligned} w_i &= b^{log(\tilde{d}_i+1)} \\ p_i &= \frac{w_i}{\sum_{v_j \in \mathcal{V}} w_j}, \quad for\ any\ v_i \in \mathcal{V} \end{aligned} \tag{7}$$

where $\tilde{d}_i$ denotes the degree of node v_i, computed based on the adjacency matrix $\hat{\mathbf{A}}$. The hyperparameter $b \in (0, 1)$ serves to control the nonlinear transformation applied to the weights, effectively balancing the relative significance among nodes. Subsequently, the transformed weights w_i undergo normalization, yielding the sampling probability p_i associated with each node.

To guarantee a uniform distribution of sampling points within the unit interval, a low-discrepancy one-dimensional point set k_i is constructed using the following approach:

$$k_t = \left\{ \frac{2t-1}{2T_1} + \lambda_t \right\} \ mod\ 1, \quad \text{for } t \in \{1, \ldots, T_1\} \tag{8}$$

where the hyperparameter T_1 denotes the number of anchor points employed to construct global relationships. In particular, a low-discrepancy point set is initially generated using the quasi-Monte Carlo method. To enhance randomness and ensure uniform coverage over the entire sampling interval, an additional random perturbation is denoted as $\lambda_t \sim \mathcal{U}(-1/2T_1, 1/2T_1)$, this strategy effectively mitigates potential systematic biases and sampling defects inherent in deterministic sampling methods. Subsequently, for each node $v_i \in \mathcal{V}$ within the graph, anchor points are sampled from a global view to capture its global relationships, with the corresponding set of indices denoted as $S_i^g = (j_1, j_2, \ldots, j_{T1})$.

To establish local relationships, we employ personalized PageRank to simulate information propagation within the graph structure, enabling a more accurate representation of local connectivity between nodes. Initially, we transform the adjacency matrix into a probability matrix $\mathbf{P}_{ij} = \tilde{\mathbf{A}}_{ij} / \sum_{k=1}^{M} \tilde{\mathbf{A}}_{ik} \quad \forall i, j \in \mathcal{V}$, which represents the probability of transitioning from node i to node j. Unlike the standard PageRank, personalized PageRank introduces a personalized restart vector $\boldsymbol{r}$, satisfies $\sum_{i=1}^{M} \mathbf{r}_i(j) = 1, \ \mathbf{r}_i(j) \geq 0$, if $\forall j \in \mathcal{V}$. This vector effectively guides the random walk process to preferentially concentrate on designated target nodes, thereby precisely uncovering nuanced local structural characteristics

Algorithm 1 GMARC

Input: HSI data $\mathcal{S}$; Number of neighbors k; Number of superpixels M; Masking rate ξ; Number of epoch I.
Output: Clustering results **R**.

```
Apply the SLIC to segment the input HSI data S into a superpixel set T.
Construct the main input graph G_1 using the superpixel centers as anchor points.
for i = 1 to I do
    Construct the masked feature matrix X̃ and the masked graph G_2 by Eq. (2).
    Re-mask the latent feature Z_2 using Eq. (5) to yield Z̃_2, and then decode it to generate X̂.
    Construct the global relationships S_i^g using Eq. (7-8), and construct the local relationships S_i^l using Eq. (9).
    Compute the similarity among distinct relationships by Eq. (10-11), yielding e_{1,i}^g, e_{1,i}^l, e_{2,i}^g, and e_{2,i}^l.
    Perform relationship awareness on each group using Eq. (13), yielding S_{11}^g, S_{11}^l, S_{12}^g, and S_{12}^l.
    Calculate the matching loss L_mat by Eq. (4).
    Calculate the reconstruction loss L_sce by Eq. (6).
    Calculate the aware loss L_awa by Eq. (14).
    Optimize the network parameters using Adam by minimizing Eq. (15).
end for
Apply the k-means to the reconstructed features X̂ to generate the final clustering results R.
return R
```

embedded within the graph. Based on the aforementioned idea, the definition of personalized PageRank can be expressed as

$$\mathbf{u}_i^{(t+1)} = \eta \mathbf{P}^T \mathbf{u}_i^{(t)} + (1-\eta)\,\mathbf{r} \tag{9}$$

where η denotes the walk probability, $\mathbf{u}_i^{(t+1)}$ represents the node feature vector at the $(t+1)$th iteration, and the final vector $\mathbf{u}$ corresponds to the converged solution of the equation.

For each node $v_i \in \mathcal{V}$, we identify the top T_2 highest scoring nodes to construct its local relationships $S_i^l = (j_1, j_2, \ldots, j_{T2})$. It ensures that the constructed local relationships exhibit a high degree of cohesion, thereby enhancing their effectiveness in subsequent computational tasks. Crucially, by strategically restricting the number of sampled global and local anchor nodes, this design guarantees that the computational complexity of the RAL module scales strictly linearly with the graph size, avoiding the quadratic bottleneck typical of full pairwise global attention computations.

2) Relationship Awareness: To learn more discriminative representations, we fully exploit the constructed multiple relationships to actively differentiate between positive and negative sample pairs. In particular, instead of relying on heuristic data augmentations, our mechanism explicitly conceptualizes the identical node across different graph views, as well as nodes sharing high structural affinity derived from the global and local anchor sets, as positive pairs. Conversely, representations of distinct nodes with divergent structural contexts are treated as negative pairs. By explicitly aligning the relational embeddings derived from these anchor sets, the model actively

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encourages positive samples to exhibit high semantic consistency while simultaneously striving to minimize spurious correlations among negative samples. For convenience in later representations, we define $\mathbf{z}_{1,i}$ and $\mathbf{z}_{2,j}$ as the ith and jth row vectors of matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$, respectively. $\langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j} \rangle$ represents the dot product of these two vectors.

To operationalize this active recognition mechanism and reveal the intrinsic connection between the relationships constructed by graph G_1 and graph G_2, we evaluate the structural contexts using cosine similarity. The metric is defined as follows:

$$\begin{aligned} \mathbf{e}_{1,i}^{g} &= \left[\langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_1} \rangle, \langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_2} \rangle, \ldots, \langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_{T_1}} \rangle \right]^T \\ \mathbf{e}_{1,i}^{l} &= \left[\langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_1} \rangle, \langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_2} \rangle, \ldots, \langle \mathbf{z}_{1,i}, \mathbf{z}_{2,j_{T_2}} \rangle \right]^T . \end{aligned} \tag{10}$$

Meanwhile, it is equally important to preserve discriminability among different relations within the graph. To this end, we apply the same approach to assess the correlations among distinct relationships embedded in the same graph structure

$$\begin{aligned} \mathbf{e}_{2,i}^{g} &= \left[\langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_1} \rangle, \langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_2} \rangle, \ldots, \langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_{T_1}} \rangle \right]^T \\ \mathbf{e}_{2,i}^{l} &= \left[\langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_1} \rangle, \langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_2} \rangle, \ldots, \langle \mathbf{z}_{2,i}, \mathbf{z}_{2,j_{T_2}} \rangle \right]^T . \end{aligned} \tag{11}$$

For each node in G_1 and G_2, we concatenate its corresponding column vectors along the rows to obtain the relational embedding $\mathbf{E}_1^g = \text{concat}(\mathbf{e}_{1,1}^g, \ldots, \mathbf{e}_{1,i}^g, \ldots, \mathbf{e}_{1,M}^g)$. Similarly, the representations of $\mathbf{E}_1^l$, $\mathbf{E}_2^g$, and $\mathbf{E}_2^l$ are obtained. To realize the active differentiation of sample pairs, we enforce semantic consistency between a node's relational embedding and its corresponding counterpart across different views, while penalizing the associations between different nodes. We partition the four existing relational embeddings into four groups, denoted as $\{\mathbf{E}_m^v, \mathbf{E}_n^v\}$, where $m, n \in \{1, 2\}$, and $v \in \{g, l\}$. We introduce the cross correlation matrix

$$\mathbf{S}_{mn}^{v} = \frac{\mathbf{E}_m^v \cdot \mathbf{E}_m^{v\,T}}{\left\| \mathbf{E}_m^v \right\|} \cdot \frac{\mathbf{E}_n^v \cdot \mathbf{E}_n^{vT}}{\left\| \mathbf{E}_n^v \right\|}. \tag{12}$$

This cross correlation matrix serves as the core of our active recognition mechanism. Its diagonal elements, representing the correlation of the same node across different relational views, are explicitly optimized toward 1. Simultaneously, the off-diagonal elements, representing the correlation between distinct nodes, are penalized toward 0. This mechanism is mathematically formulated as the following relational correlation objective for each group:

$$R_{cor}(S) = \frac{1}{M} \sum_{i=1}^{M} (S_{ii} - 1)^2 + \frac{1}{M(M-1)} \sum_{i \neq j} S_{ij}^2. \tag{13}$$

Based on the above equation, we can derive the aware loss $\mathcal{L}_{awa}$

$$\mathcal{L}_{\text{awa}} = R_{cor}\left(S_{11}^{g}\right) + R_{cor}\left(S_{12}^{g}\right) + R_{cor}\left(S_{11}^{l}\right) + R_{cor}\left(S_{12}^{l}\right). \tag{14}$$

E. Loss Function

The loss function in GMARC is specifically designed to effectively guide the learning of the model by aligning the latent representations derived from masked node encodings, minimizing the discrepancy between features of the original graph and its reconstructed counterpart, and capturing intrinsic correlations among multiple relationships. In particular, the proposed loss function comprises three distinct components: matching loss, reconstruction loss, and aware loss.

In summary, the final overall loss function of the framework is defined as

$$\mathcal{L} = \mathcal{L}_{\text{sce}} + \alpha_1 \mathcal{L}_{\text{mat}} + \alpha_2 \mathcal{L}_{\text{awa}} \tag{15}$$

where α_1 and α_2 are tradeoff parameters among the three loss functions.

IV. Experiments

A. Datasets

To assess the performance of our model, we conduct experiments on three publicly available HSI datasets: Xu Zhou, Salinas, and Trento.

1) *Xu Zhou:* This dataset was acquired via the airborne HySpex sensor over Xu Zhou, Jiangsu. It consists of 436 spectral bands, offering high spatial resolution and extensive spectral information.
2) *Salinas:* The second dataset was acquired by the AVIRIS sensor in the Salinas Valley, CA, USA. It has a spatial resolution of 3.7 m and includes 16 distinct land cover types. After excluding the interference bands, the dataset consists of 204 spectral bands. The image has dimensions of 512 × 217 pixels, with 54 129 labeled pixels remaining after background removal.
3) *Trento:* The third dataset was acquired from the Trento region in Italy, containing a total of 490 000 pixels, of which 459 786 are background pixels and 30 214 are labeled pixels. This dataset comprises seven land cover types and has a spatial resolution of 1.2 m. It spans both visible and near-infrared spectral bands, with 128 bands in total.

B. Experimental Setup

To comprehensively evaluate the performance and computational efficiency of our proposed framework, we systematically compare it with thirteen advanced benchmark methods: k-means [59], SSC [60], MS^{2}A-Net [61], DGAE [62], NCSC [63], GraphMAE [39], GraphMAE2 [40], DSCRLE [64], EMVCC [57], SDST [55], S^{2}GCL [29], MHGC [65], and SCGC [66]. These comparative benchmarks encompass traditional and subspace clustering methods, including k-means, SSC, and NCSC; spectral correlation and Transformer architectures, such as MS^{2}A-Net, SDST, and DSCRLE; as well as advanced GNN and contrastive learning frameworks, notably DGAE, GraphMAE, GraphMAE2, S^{2}GCL, EMVCC, MHGC, and SCGC.

All experiments are executed on a workstation with 64 GB RAM and an RTX 3090 GPU. To evaluate computational complexity, the total training and test times are recorded alongside clustering metrics [67], [68]. The number of iterations was set to 100 epochs to ensure sufficient convergence. In addition, to mitigate randomness, each reported performance metric was averaged over ten independent repeated experiments. During

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TABLE I
EXPERIMENTAL CLUSTERING RESULTS OF COMPARED APPROACHES ACROSS XU ZHOU, SALINAS, AND TRENTO DATASETS

Dataset	Metric	k-means	SSC	MS^{2}A-Net	DGAE	NCSC	GraphMAE	GraphMAE2	DSCRLE	EMVCC	SDST	S^{2}GCL	MHGC	SCGC	GMARC
Xu Zhou	ACC	0.4042	0.6974	0.6261	0.6761	0.7403	0.7396	0.7254	0.7106	0.7506	0.6700	0.7304	0.7612	0.6169	**0.7968**
	Kappa	0.2958	0.6323	0.5377	0.5993	0.6833	0.6779	0.6604	0.6496	0.6975	0.5772	0.6702	0.6944	0.4930	**0.7488**
	NMI	0.4240	0.5758	0.5821	0.6105	0.7239	0.6555	0.6591	0.6228	0.7323	0.6282	**0.7456**	0.6869	0.4920	0.7357
	ARI	0.2660	0.5788	0.5528	0.6136	0.6795	0.7036	0.7020	0.5674	0.6166	0.6015	0.6835	0.7208	0.4745	**0.7417**
	Purity	0.6500	0.7430	0.6749	0.7676	0.7923	0.7948	0.7962	0.7589	0.8110	0.7045	0.8056	0.7619	0.6641	**0.8455**
	Time (s)	14.53	1684.21	473	117	420	82.35	88.62	538	275.46	1842	41.28	150.91	136.37	39
Salinas	ACC	0.6111	0.6546	0.6463	0.5950	0.7497	0.6977	0.6930	0.5713	0.6796	0.7433	0.7623	0.6322	0.6558	**0.8138**
	Kappa	0.5758	0.6234	0.6031	0.5560	0.7207	0.6653	0.6612	0.5289	0.6461	0.7130	0.7368	0.5887	0.6205	**0.7941**
	NMI	0.7124	0.7647	0.7447	0.7089	0.8169	0.7092	0.7156	0.6086	0.7260	0.8011	**0.8399**	0.7439	0.7298	0.8037
	ARI	0.4820	0.5802	0.5995	0.4787	0.6651	0.5897	0.5883	0.3594	0.5302	0.6186	0.6889	0.5638	0.5201	**0.7215**
	Purity	0.7030	0.7537	0.6630	0.6958	0.7638	0.7215	0.7326	0.6069	0.7086	0.7764	0.7801	0.6668	0.7268	**0.8421**
	Time (s)	11.24	2315.76	488	46	500	95.18	102.45	125	342.15	986	48.33	273.02	156.46	58
Trento	ACC	0.6281	0.7659	0.8360	0.7985	0.6685	0.8607	0.8306	0.7711	0.8003	0.8296	0.7239	0.8036	0.7244	**0.9085**
	Kappa	0.5296	0.6965	0.7814	0.7300	0.5823	0.8159	0.7799	0.6904	0.7446	0.7655	0.6451	0.7446	0.6483	**0.8781**
	NMI	0.4618	0.6692	0.7624	0.6588	0.6307	0.8139	0.7834	0.6569	0.7139	0.7192	0.6828	0.7373	0.5918	**0.8639**
	ARI	0.3426	0.7170	0.8128	0.7065	0.5806	0.8561	0.8008	0.7031	0.7067	0.7490	0.6769	0.7693	0.4973	**0.9266**
	Purity	0.6480	0.8295	0.8419	0.8092	0.7748	0.8944	0.8938	0.8011	0.8803	0.8296	0.8062	0.8242	0.7321	**0.9174**
	Time (s)	4.85	624.39	106	39	435	45.22	48.71	124	145.82	552	28.56	82.03	47.73	36

model optimization, we employed the Adam optimizer with a stepwise learning rate decay strategy, reducing the initial learning rate by a factor of 0.1 every ten epochs to ensure convergence stability. Moreover, all graph encoders and decoders in our GMARC model use a two-layer GCN, with hidden and output layer dimensions set to 128 and 256, respectively. In addition, the scaling factor γ in the reconstruction loss in (6) was set to 2, while the personalized PageRank walk probability η within the RAL module was fixed at 0.85. To guarantee a fair and rigorous comparison with existing baseline methods, the comparative experiments were conducted using their publicly available source codes. Rather than relying on default configurations, we carefully retuned all baselines on the evaluated datasets. In particular, a consistent hyperparameter tuning budget was strictly applied across all methods, with baseline search spaces defined by their original literature. It should be noted that since our framework focuses exclusively on unsupervised clustering, the entire dataset is utilized for label-free representation learning and subsequent evaluation. To quantitatively evaluate the clustering performance, we utilized five standard academic indicators: accuracy (ACC), the Kappa coefficient (Kappa), normalized mutual information (NMI), the adjusted rand index (ARI), and purity [69], [70], [71], [72].

C. Comparison With SOTA Methods

Table I presents a quantitative comparison of GMARC against baseline methods, encompassing both clustering accuracy metrics and empirical training and test times. For each clustering metric, the best performance is highlighted in bold, while the second-best value is underlined. The results show that the proposed method improves the ACC metric by 3.56%, 5.15%, and 4.78% compared to the second-best method on the Xu Zhou, Salinas, and Trento datasets, respectively. Furthermore, GMARC achieves the best results across all metrics for the Trento dataset, except the NMI metric for the Xu Zhou and Salinas datasets, where it does not reach the highest value, trailing slightly behind S^{2}GCL. This inconsistency stems from the distinct mathematical properties of the metrics. ACC and purity evaluate point-to-point correctness, where GMARC excels by capturing dominant spatial–spectral consistencies through superpixels and RAL. Conversely, NMI measures the information-theoretic alignment of cluster distributions. While GMARC's masked reconstruction robustly extracts deep structures, it may induce slight oversmoothing on minor classes or complex boundaries. This marginally impacts the entropy alignment measured by NMI, giving a slight edge to contrastive methods like S^{2}GCL that explicitly maximize interclass distances without generative smoothing. These results provide strong evidence of the effectiveness of our method.

To visually demonstrate the advantages of the proposed method, we present cluster visualization results for 13 different methods on the Xu Zhou, Salinas, and Trento datasets in Figs. 3–5. The results show that traditional clustering approaches [59], [60] often fail to adequately leverage the rich spectral information inherent in HSI, resulting in limited discriminative feature representations and consequently constrained clustering performance. In comparison, MS^{2}A-Net achieves smoother clustering results by constructing a spectral correlation network. However, its scalability remains limited for large-scale HSI datasets, primarily due to the lack of guidance provided by superpixel segmentation and the absence of constraint mechanisms during the gradient optimization stage. NCSC integrates subspace clustering and superpixel segmentation techniques and utilizes a lightweight convolutional neural network to optimize the features, thereby significantly enhancing the clustering performance.

Graph-based clustering methods have significantly enhanced the effective utilization of HSI by aggregating semantically informative neighborhood features. Visualization and data representation analyses indicate that these graph-based approaches notably outperform all baseline methods. In particular, DGAE, GraphMAE, and GraphMAE2 leverage the architecture of graph autoencoders, where GraphMAE and GraphMAE2 further incorporate masking strategies. This integration substantially improves the model's ability to capture and interpret spectral characteristics, resulting in more stable and spectrally informative graph representations. On the contrary, S^{2}GCL utilizes a graph encoder combined with a

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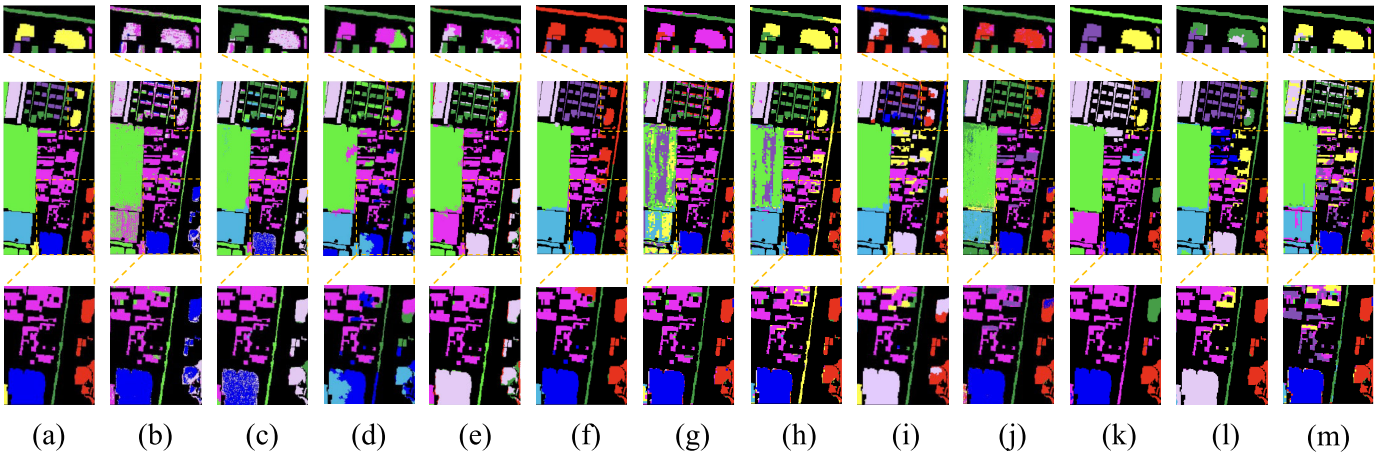


Fig. 3. Clustering visualization results on Xu Zhou dataset. (a) Ground truth. (b) *k*-means. (c) SSC. (d) MS^{2}A-Net. (e) DGAE. (f) NCSC. (g) GraphMAE. (h) GraphMAE2. (i) DSCRLE. (j) EMVCC. (k) SDST. (l) S^{2}GCL. (m) Ours.

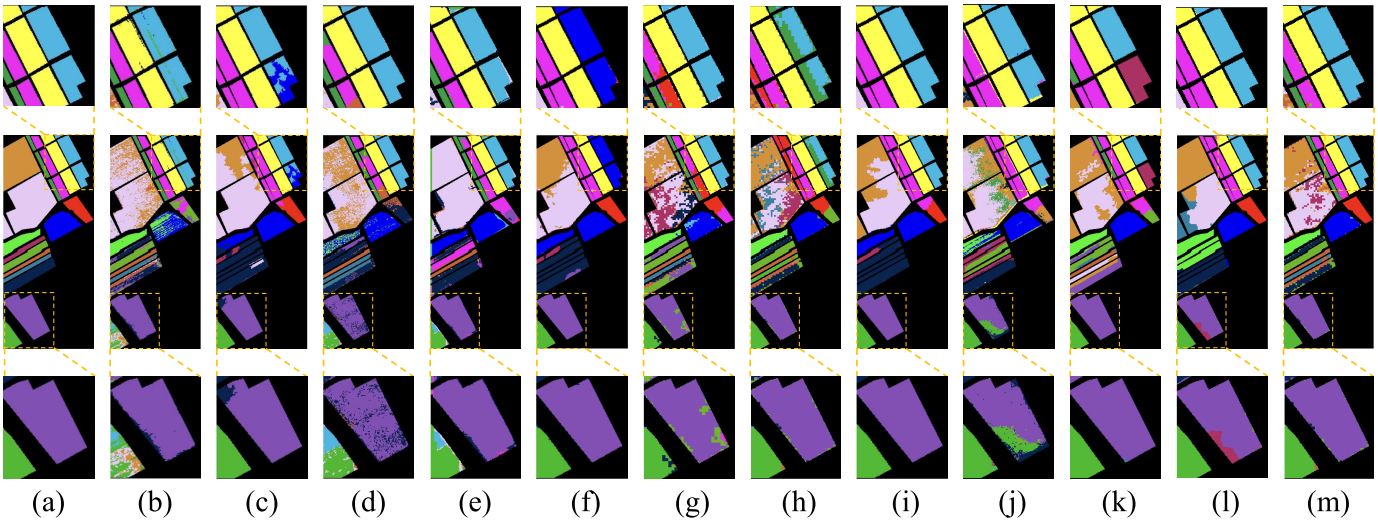


Fig. 4. Clustering visualization results on the Salinas dataset. (a) Ground truth. (b) *k*-means. (c) SSC. (d) MS^{2}A-Net. (e) DGAE. (f) NCSC. (g) GraphMAE. (h) GraphMAE2. (i) DSCRLE. (j) EMVCC. (k) SDST. (l) S^{2}GCL. (m) Ours.

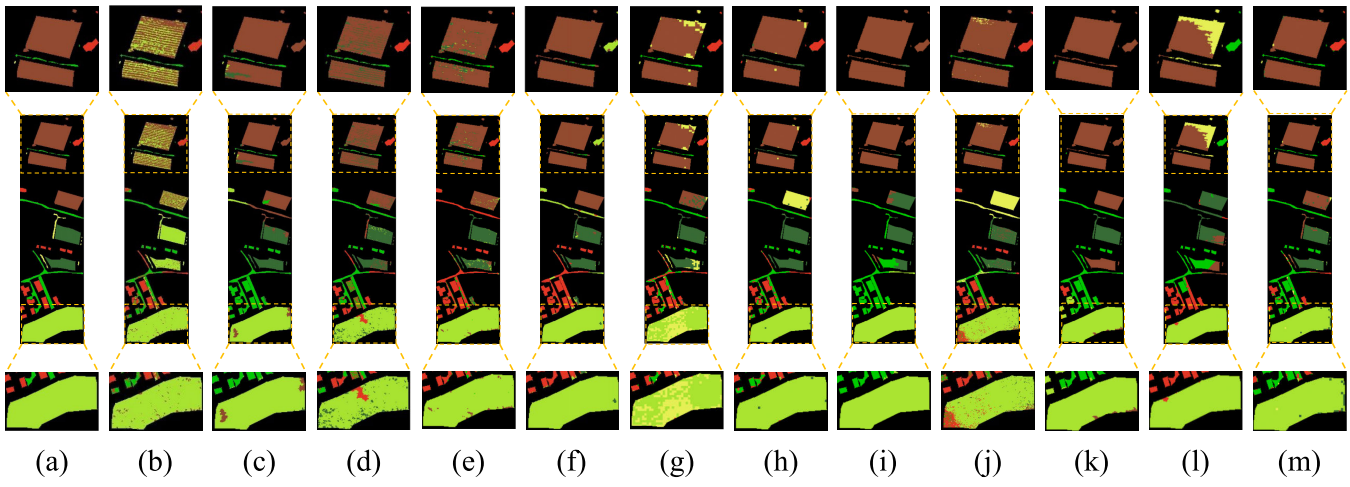


Fig. 5. Clustering visualization results on Trento dataset. (a) Ground truth. (b) *k*-means. (c) SSC. (d) MS^{2}A-Net. (e) DGAE. (f) NCSC. (g) GraphMAE. (h) GraphMAE2. (i) DSCRLE. (j) EMVCC. (k) SDST. (l) S^{2}GCL. (m) Ours.

contrastive learning mechanism, thereby further advancing clustering performance. However, their reliance on complex negative sampling or augmentation strategies may inadvertently distort the intrinsic graph topology, thereby limiting their

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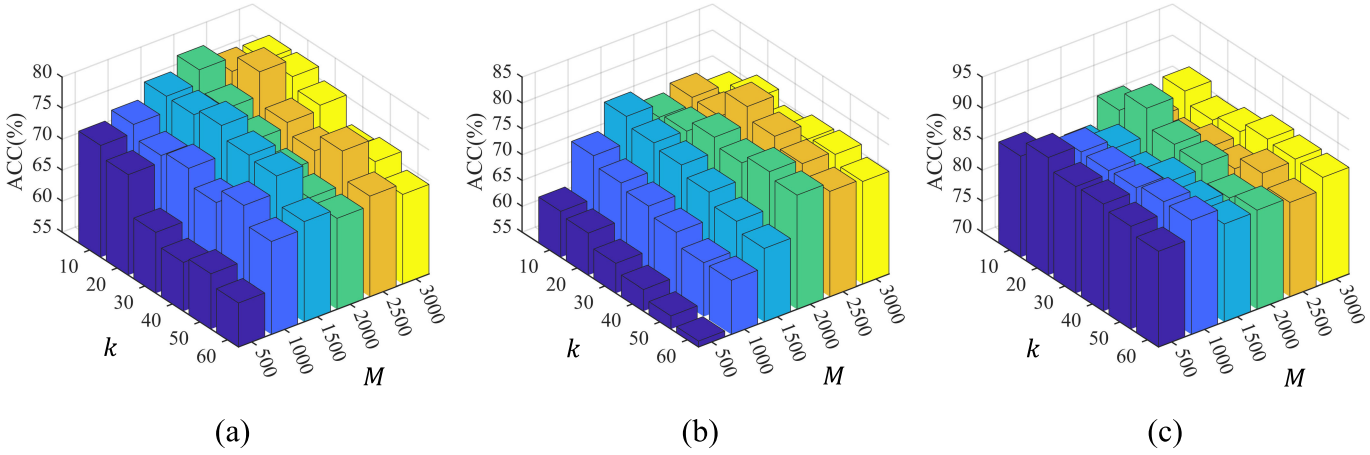


Fig. 6. 3-D comparative evaluation of clustering performance for different values of k and M on (a) *Xu Zhou*, (b) *Salinas*, and (c) *Trento*.

optimal performance on complex HSI datasets. Motivated by the strengths of these methods, this study proposes a novel GMARC framework, which integrates MGAE with RAL. GMARC effectively synthesizes the advantages of the aforementioned techniques and achieves superior clustering results across three benchmark datasets. Furthermore, as evidenced by the empirical execution times reported in Table I, GMARC maintains competitive computational efficiency, which is supported by its strictly linear computational scalability with respect to the superpixel count.

D. Parameter Settings and Analysis

1) Sensitivity Analysis of the Superpixel Number M: The number of superpixels, denoted as M, determines the granularity of the image segmentation process. An optimal selection of M is highly dependent on the spatial resolution and the landscape complexity of the dataset. To assess the influence of M, we conducted experiments with varying values chosen from the set $S = \{500, 1000, 1500, 2000, 2500, 3000\}$, and the detailed results are presented in Fig. 6. The findings indicate that smaller values of M result in excessively coarse segmentation. In fragmented scenarios, this causes individual superpixels to encompass multiple distinct land cover types, thereby failing to capture fine-grained boundaries and negatively affecting clustering accuracy. Conversely, excessively large values of M lead to superpixels that contain redundant information, which impairs the model's ability to extract broader contextual details and increases computational overhead. Based on the spatial complexity of the scenes, the optimal number of superpixels selected for the Xu Zhou, Salinas, and Trento datasets is 2500, 2500, and 2000, respectively.

2) Sensitivity Analysis of the Nearest Neighbor k: The choice of the k value for nearest neighbors plays a critical role in constructing the initial graph topology. To examine its influence, we performed experiments selecting k values ranging from 10 to 60 with a step size of 10, as illustrated in Fig. 6. The optimal k is strongly correlated with the spatial homogeneity of the dataset. When smaller k values are chosen, the resulting graph structure often lacks sufficient high-dimensional feature interaction, leading to reduced model robustness. However, in datasets with high spatial heterogeneity, a larger k introduces indistinguishable neighbor noise from different classes, causing a severe oversmoothing effect. For datasets with larger contiguous homogeneous regions, the model can tolerate and benefit from a larger k. In our experiments, we selected optimal k values of 20, 30, and 20 for the Xu Zhou, Salinas, and Trento datasets, respectively.

3) Sensitivity Analysis of the Tradeoff Parameters α_1 and α_2: Fig. 7 illustrates the variations in ACC across the datasets as α_1 and α_2 vary from 0.01 to 1000. The Xu Zhou dataset exhibits more pronounced fluctuations compared to Salinas and Trento, likely due to substantial inherent noise interference. When α_1 is set to a low value, clustering performance is generally suboptimal; however, as α_1 increases, results improve, underscoring the importance of matching alignment in guiding the autoencoder's reconstruction. Regarding α_2, the best parameter values across all datasets are concentrated in the range $\alpha_2 \geq 1$. This indicates that the proposed relational awareness learning module effectively discriminates between positive and negative sample pairs, confirming its efficacy.

4) Sensitivity Analysis of the Masking Rate ξ: Fig. 8 illustrates the variation in model performance across different masking rates. The impact of ξ is directly linked to the information redundancy of the HSI data. In most cases, lower masking rates fail to create sufficiently challenging reconstruction tasks, resulting in superficial feature learning. For the Xu Zhou and Trento datasets, which feature more complex and noisy spatial distributions, the optimal clustering performance was achieved with a masking rate of 30%. In stark contrast, for the Salinas dataset, a much higher masking rate of 70% yielded the best results. This phenomenon is driven by the high homogeneity and strong information redundancy within the Salinas image. Even at a high masking rate, the missing features can be accurately inferred from a small subset of unmasked neighboring nodes, effectively pushing the model to learn a more robust underlying representation.

E. Ablation Study

To investigate the specific contributions of individual components within the proposed GMARC framework, we

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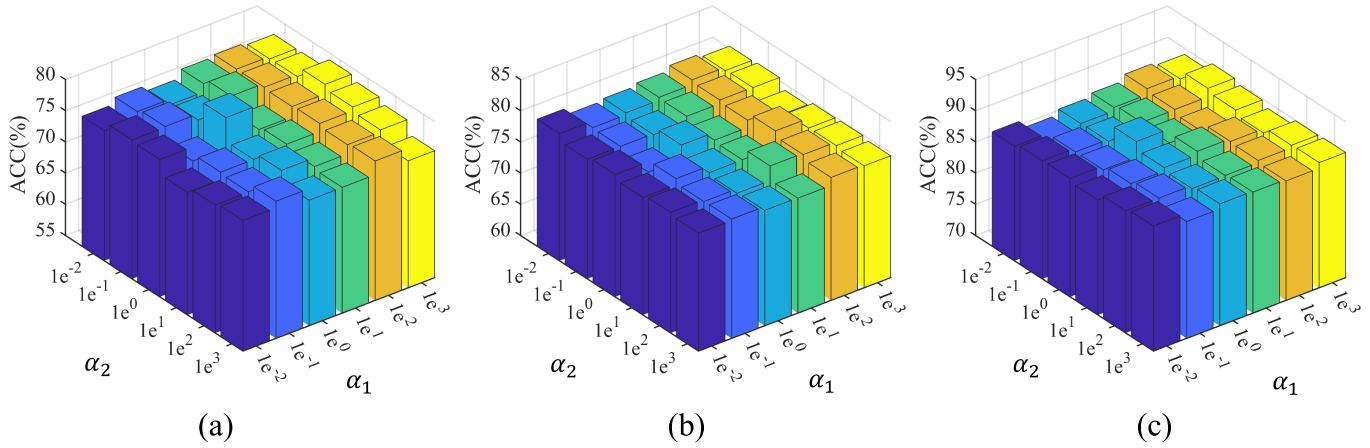


Fig. 7. 3-D comparative evaluation of clustering performance for different values of α_1 and α_2 on (a) *Xu Zhou*, (b) *Salinas*, and (c) *Trento*.

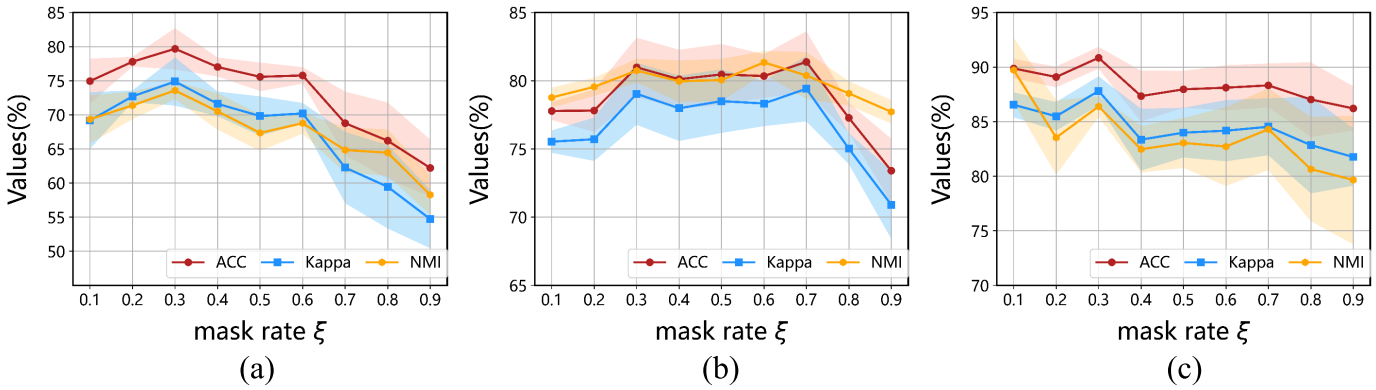


Fig. 8. Comparative evaluation of clustering performance for different mask rates ξ on (a) *Xu Zhou*, (b) *Salinas*, and (c) *Trento*.

TABLE II
ABLATION STUDIES ON RECONSTRUCTED FEATURES (REC.), RAL (RAL.), AND MATCH ALIGNMENT (MAT.) FOR GMARC

No.	REC.	RAL.	MAT.	Xu Zhou	Salinas	Trento
1	✓		✓	0.7483	0.7804	0.8731
2		✓	✓	0.7640	0.7851	0.8985
3	✓	✓		0.7717	0.7919	0.8804
4	✓	✓	✓	**0.7968**	**0.8138**	**0.9085**

conducted ablation studies on three key modules: reconstructed features, RAL, and match alignment. The clustering performances were quantitatively evaluated on three benchmark HSI datasets: Xu Zhou, Salinas, and Trento. As presented in Table II, each module positively impacts performance, with their integration yielding the highest accuracy as anticipated.

Notably, the RAL module exhibits the most substantial contribution, achieving an average clustering accuracy improvement of approximately 3.91% across all three datasets. This module effectively enhances the discriminative capability of the model regarding characteristics by promoting the consistency between representations of positive sample pairs and suppressing redundant correlations among negative samples. In addition, the clustering results of scheme No. 2 further demonstrate that the masking operation compels the autoencoder to mine deeper underlying feature information within HSI, thereby enabling both the encoder and decoder to achieve more refined and comprehensive feature representations during reconstruction. Consequently, the model exhibits enhanced feature inference capabilities. Furthermore, the match alignment module introduces alignment constraints on masked-node latent features generated by the online–target encoder, effectively guiding the model to capture and reconstruct more intricate and challenging feature details.

F. Discussion and Limitations

To provide a comprehensive evaluation, we discuss the advantages and limitations of GMARC relative to existing methodologies. Unlike contrastive learning methods [29], [57] that rely on complex data augmentations, potentially distorting the intrinsic graph topology, GMARC utilizes a generative masked reconstruction task. This preserves structural integrity and makes the method more robust to noise. Furthermore, compared to standard graph autoencoders like DGAE, our RAL module explicitly models global Non-IID relationships to prevent the representation collapse often seen in simple reconstruction tasks. However, the incorporation of the RAL module and the dual masking strategy increases the computational complexity of the training phase compared to standard GCN baselines. Nevertheless, GMARC avoids the

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quadratic FLOPs and parameter expansion typical of global attention mechanisms, achieving a strictly linear theoretical complexity of $\mathcal{O}\left(M \cdot k\sum_{i=1}^{L} d_i d_{i-1} + M \cdot T(\hat{d} + T)\right)$. This ensures optimal training and test efficiency compared to the cubic complexity of traditional subspace clustering [60].

Regarding limitations, GMARC constructs the initial graph based on superpixels generated by the SLIC algorithm. Consequently, the final clustering performance is partially dependent on the quality of this segmentation. As shown in Fig. 5, an inappropriate number of superpixels M can lead to under segmentation or over segmentation, which limits the upper bound of clustering accuracy.

V. Conclusion

In this article, we propose a novel GMARC model for HSI clustering, aiming to mitigate the issues of representation collapse during model training and structural distortion induced by conventional data augmentation techniques. The proposed GMARC effectively leverages intrinsic characteristics and explores the high-dimensional topological structure inherent in HSI data. In particular, a graph masking mechanism is introduced to facilitate robust learning of stable representations from input graphs enriched with spectral information. Furthermore, we devise an innovative data augmentation strategy tailored to exploit the high-dimensional graph attributes, thereby enabling more precise extraction of discriminative features. Extensive experiments conducted on three widely adopted HSI benchmark datasets demonstrate that the GMARC model outperforms existing state-of-the-art baselines. However, the performance of this framework remains partially bounded by the quality of the initial superpixel segmentation and its sensitivity to domain-specific masking rates. To mitigate these dependencies and explore richer feature spaces, future work will further investigate the fusion of multimodal information to improve clustering efficacy.

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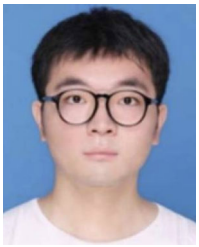
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